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Assignment -1

1. Data Link Layer

a) The Data link layer is responsible for a transferring data between two directly connected devices over a physical network.

Functions:

1) Framing: Divides data into frames for transmission
2) Physical addressing: uses MAC addresses to identify devices on a local network.
3) Error Detection: Detects transmission Errors using Methods such as CRC.
4) Flow Control: Helps prevent Sender from overwhelming a reciever when supported by the link protocol

2. Transport Layer

The transport layer provides communication between application on different device

Segmentation: Divides data into Smaller Segments port Addressing: Identifies application. Error Recovery: Prevents receives. ~~overload~~ reliable D

Reliable Delivery : TCP provide ordered and reliable data delivery

b)

Connection Oriented	Connectionless
Establishe a connection first	Do not require connection Setup.
TCP provides reliable delivery	UDP does not guarantee delivery
Ex: File transfering TCP	Ex: DNS queries using UDP.

2. a) Encapsulation and Recapsulation

Encapsulation: Each osi layer adds its header as data moves from sender to reciever's network.

Recapsulation: the reciever removes header layer by layer to Obtain the original data.

Order : data :→ Segment → packeet → Frame → Bits.

c) Network layer

The network layer performs routing between networks. It uses logical IP addresses to identifies destination and forwards packet through routers.

(3) a) Classful Addressing and CIDR

classful addressing uses fixed networks size. It works IP adresses and provides limited flexibility.

CIDR uses flexibility. Prefix length such as 126. It reduces address wastage and supports route aggregation.

b) Subnet calculation

Given IP: 192.168.25.139.

Mask: 255.255.255.192(126)

Network ID: 192:168.25.128.

Broadcast: 192:168.25.191

First usable: 192.168.25.129

Last usable: 192:168.25.190

Usable hosts: 62.

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a) Subnet mask

Given : 172.16.0.0/16 at least 6 subnets

$2^3 = 8$ subnets

Borrowed bits = 3

New CIDR = /19

Subnet mask : 255.255.224.0

b) Subnet and hosts.

Total subnets : $2^3 = 8$

Host bits : $32 - 19 = 13$.

usable hosts : $2^{13} - 2 = 8190$

c) First two subnets

Details	Subnet - 1	Subnet 2
Network ID	172.16.0.0	172.16.32.0
First Usable	172.16.0.1	172.16.32.1
last usable	172.16.31.254	172.16.63.254
Broadcast	172.16.31.255	172.16.63.255

(5) a) Statics Vs Dynamic Routins

- Static routing: Route are configuard normally. Suitable for small, stable nehoorks
- Dynamic routing: Route are learned automatically using protocols such as RIP and OSPF
- Static routing uses fixed resources while dynamic routing adopts to network change

b) Static route Configuration

Router > Enable
Route # configure terminal
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)# end.
Route # show ip route.